UNIT 2 · TOPIC 9(A)

Hydraulic Controls

Syllabus topic: Pneumatic and hydraulic controls

Principle

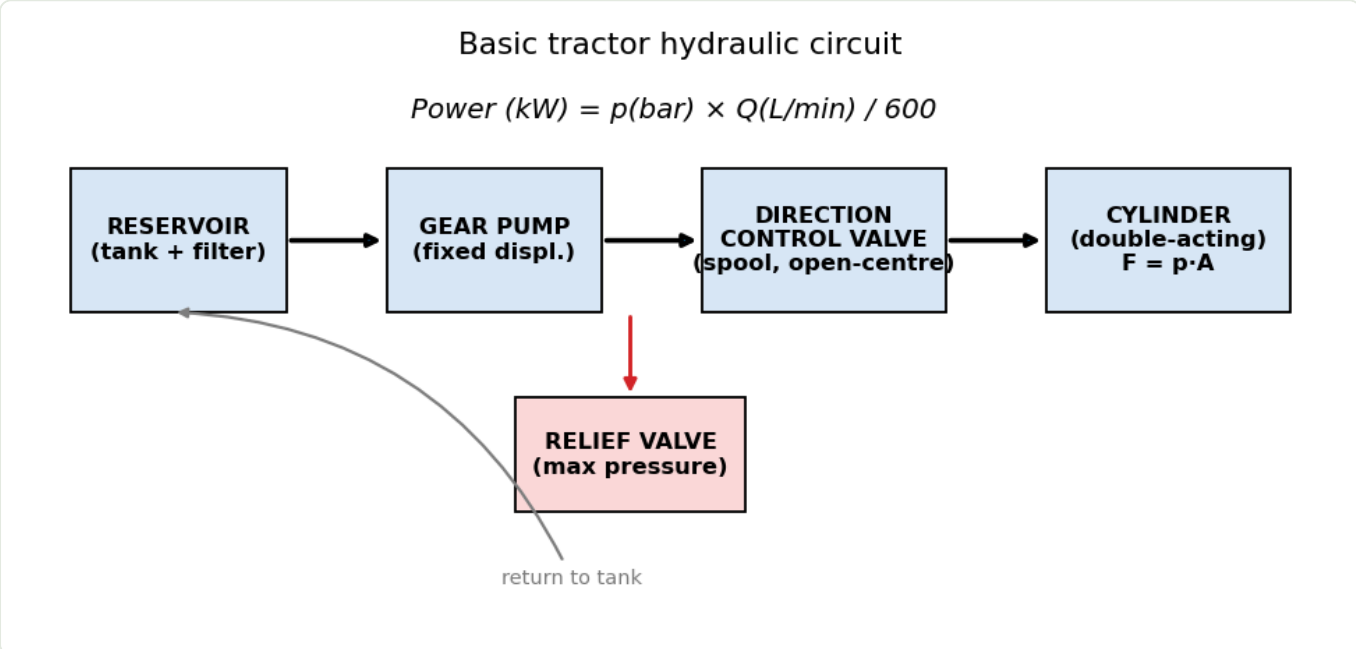
- **Pascal's law:** pressure applied to a confined fluid is transmitted equally in all directions.
- Oil is practically incompressible → stiff, precise, high force-density actuation.

$$F = p \times A$$

Example. 200 bar (20 MPa) acting on a 50-cm² piston → F = 20×10^{6} × 50×10^{-4} = **100 kN**.

Components, in order of oil flow

1. **Reservoir** with filter and breather.
2. **Pump** — tractors use fixed-displacement external gear pumps (cheap, robust); piston pumps for high pressure.
3. **Pressure relief valve** — limits maximum system pressure and protects the circuit.
4. **Directional control valve** — spool type; directs oil to either side of the cylinder.
5. **Actuator** — single-acting cylinder (oil lifts, gravity returns — e.g., 3-point hitch ram), double-acting cylinder (oil-powered both ways — loaders, steering), or hydraulic motor (continuous rotation — orbital motors on seed-drill fans, combine headers).



Basic tractor hydraulic circuit

$$\text{Hydraulic power } P(\text{kW}) = p(\text{bar}) \times Q(\text{L/min})/600$$

Example. 180 bar at 40 L/min → **12 kW**.

System types

System	How it works
Open-centre	A fixed-displacement pump runs continuously; in neutral the valve centre is OPEN and flow returns freely to tank at low pressure. Simple — standard on most agricultural tractors.
Closed-centre	The valve centre is closed in neutral; a variable-displacement pump maintains pressure and supplies flow only on demand. Load-sensing versions match pump flow and pressure to the heaviest load — efficient, used on larger modern tractors.

Applications

- Three-point hitch raise/lower with automatic **position control** (implement held at a set height) and **draft control** (hitch raised/lowered to hold the set draft, sensed by top-link or lower-link load sensing) — details in Unit 4.
- Remote (auxiliary) cylinders for trailed implements.
- Hydrostatic transmission (pump + motor) on combines and self-propelled machines.
- Hydraulic power steering.
- Loader, dozer and reversible-plough turnover rams.

Practice MCQs

Q181. Hydraulic systems work on:

(a) Bernoulli's theorem (b) Pascal's law
(c) Archimedes' principle (d) Boyle's law

Q182. Oil at 200 bar acts on a 50-cm² piston. The force developed is:

(a) 10 kN (b) 50 kN
(c) 100 kN (d) 200 kN

Q183. Hydraulic power in kW with p in bar and Q in L/min is:

(a) pQ/60 (b) pQ/367
(c) pQ/600 (d) pQ/1000

Q184. The hydraulic pump almost universally used on agricultural tractors is:

(a) Centrifugal (b) External gear
(c) Diaphragm (d) Screw

Q185. In an OPEN-centre hydraulic system, in neutral the pump flow:

(a) Is blocked (b) Returns freely to tank at low pressure
(c) Goes to the cylinder (d) Stops

Q186. A closed-centre load-sensing system uses a:

(a) Fixed-displacement pump running at relief pressure (b) Variable-displacement pump supplying flow on demand
(c) Hand pump (d) Air compressor

Q187. The relief valve in a hydraulic circuit:

(a) Directs flow (b) Limits maximum system pressure
(c) Filters oil (d) Cools oil

Q188. A single-acting hydraulic cylinder is lowered by:

(a) Oil pressure
(b) Gravity/implement weight
(c) Spring only
(d) Air

Answers

Q181 — (b) Pascal's law

Q182 — (c) 100 kN — *$F = pA = 200\times10^{5}$ Pa × 50×10^{-4} m² = 100,000 N = 100 kN — hydraulic force multiplication.*

Q183 — (c) pQ/600 — *P(kW) = p(bar)×Q(L/min)/600; e.g., 180 bar × 40 L/min → 12 kW.*

Q184 — (b) External gear

Q185 — (b) Returns freely to tank at low pressure

Q186 — (b) Variable-displacement pump supplying flow on demand

Q187 — (b) Limits maximum system pressure

Q188 — (b) Gravity/implement weight

Sources: Kepner et al. (hydraulic principles); Drone Rules 2021 (Ministry of Civil Aviation); MoA&FW SOPs for drone application of pesticides & nutrients; SMAM drone guidelines & PIB releases (Kisan Drone, Namo Drone Didi — verified July 2026); ISO 11783 (ISOBUS); NPTEL lectures on mechatronics & precision agriculture.